

```
#include <stdio.h>
```

```
#include <pthread.h>
```

```
#include <semaphore.h>
```

```
Sem_t mutex;
```

```
Int readcount = 0;
```

```
Int data = 10;
```

```
Void *reader(void *arg)
```

```
{
```

```
    Sem_wait(&mutex);
```

```
    Readcount++;
```

```
    Printf("Reader %d is reading data = %d\n", *(int *)arg, data);
```

```
    Readcount--;
```

```
    Sem_post(&mutex);
```

```
    Return NULL;
```

```
}
```

```
Void *writer(void *arg)
```

```
{
```

```
    Sem_wait(&mutex);
```

```
    Data += 5;
```

```
Printf("Writer is writing data = %d\n", data);
```

```
Sem_post(&mutex);
```

```
Return NULL;
```

```
}
```

```
Int main()
```

```
{
```

```
    Pthread_t r1, r2, w1;
```

```
    Int a = 1, b = 2;
```

```
    Sem_init(&mutex, 0, 1);
```

```
    Pthread_create(&r1, NULL, reader, &a);
```

```
    Pthread_create(&r2, NULL, reader, &b);
```

```
    Pthread_create(&w1, NULL, writer, NULL);
```

```
    Pthread_join(r1, NULL);
```

```
    Pthread_join(r2, NULL);
```

```
    Pthread_join(w1, NULL);
```

```
    Sem_destroy(&mutex);
```

```
    Return 0;
```

```
}
```